

- 9 (a) (i) point X shown correctly B1 [1]
- (ii) op-amp has very large / infinite gain M1
 non-inverting input is at earth (potential) / earthed / at 0 V M1
 if amplifier is not to saturate, inverting input must be (almost)
 at earth potential / 0 (V) same potential as inverting input A1 [3]
- (b) (i) total input resistance = $1.2 \text{ k}\Omega$ C1
 (amplifier) gain $(= -4.2 / 1.2) = -3.5$ C1
 (voltmeter) reading $= -3.5 \times -1.5$
 $= 5.25 \text{ V}$ A1 [3]
 (total disregard of signs or incorrect sign in answer, max 2 marks)
- (ii) (less bright so) resistance of LDR increases M1
 (amplifier) gain decreases M1
 (voltmeter) reading decreases A1 [3]

Page 5	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2010	9702	41

10 (a) X-ray taken of slice / plane / section

P1